

Ilídio Costa

Masters in Material Engineering

+351 925 870 614 ilidiomibritocosta@gmail.com

<https://ilidio-costa.github.io/>

Nationality: Portuguese



EXPERIENCE

FEUP & CATEC, Porto, Portugal — *Master's Dissertation*

February 2025 - October 2025

Titled "Parameter Optimisation of Laser Powder Bed Fusion of NiTi", my dissertation focused on the development of NiTi for L-PBF.

- Developed and executed a Design of Experiments (DoE) to minimize the number of samples, utilizing Taguchi and Response Surface methodologies, to efficiently optimize L-PBF parameters for NiTi.
- Analyzed NiTi powder feedstocks using Optical Microscopy (OM), Scanning Electron Microscopy (SEM), and particle size distribution analysis via ImageJ and custom Python code. And evaluate their compatibility with the L-PBF process.
- Conducted comprehensive analysis of manufactured samples to validate the best parameters using a wide range of advanced techniques:: SEM, EDS, EBSD (TSL OIM software), XRF, XRD, (performing phase indexing with pymatgen) and density measurement via the Archimedes method and micrograph threshold analysis.
- Minimised porosity to less than 0.1% whilst indicating future processing parameters. Statistically significant variations in chemical composition have been identified across different thickness sections.

Porto Space Team, Porto, Portugal — *Advisor & Propulsion Department Leader, INVICTUS Project*

October 2024 - October 2025

- Supervising a Bachelors Dissertation Project, titled "OTIMIZAÇÃO TOPOLÓGICA DO SUPORTE DE AVIONIC BAY" (Topological Optimisation of the Avionic Bay Support), planning the activities and teaching concepts of DfAM, material selection and topological optimisation.
- Facilitating knowledge transfer by documenting best practices protocols in English and providing historical context for key engineering decisions for INVICTUS II, a hybrid propulsion rocket which competed at European Rocketry Challenge (Euroc) 25.
- Founder of Project HyProSim (Hybrid Propulsion Simulator), awarded best poster presentation at the 1st Space Education Summit with the title "Predictive Modelling of Hybrid Rocket Engine Hot-Fire Tests".
- Coordinated a multidisciplinary team to develop the hybrid propulsion motor for INVICTUS II by scheduling weekly meetings, Gantt charts, and writing in depth protocol, literature reviews, reports and workshop in English, and report progress in weekly meetings.
- Develop analytical models in python to examine Hot Fire & Cold Flow data and researching and understanding concepts from fluid dynamics to combustion.
- Oversee the annual budget for the Propulsion Department through planning of activities, previous spendings and market analysis, and contact sponsor.

Job Related Abilities

Assembly, Integration & Testing (AIT)

- Hands-on experience coordinating the development and integration of the hybrid propulsion motor for the INVICTUS II rocket.
- Analyzed Hot Fire and Cold Flow data using analytical models; utilized Ishikawa diagrams to pinpoint causes of thrust deviations.
- Executed tensile, hardness, and fatigue testing to validate material performance, ensuring structural integrity for flight components.
- Proven ability to troubleshoot process failures using critical thinking and root cause analysis tools

Mechanical Operations Manufacturing

- Hands-on with L-PBF machine Prima Sharp 150 and operations such as Facing, Cutting, Drilling, Facing, and Polishing.
- Advanced user of Siemens NX and nTop for Design for Additive Manufacturing (DfAM) and Topology Optimization.
- Defined and executed heat treatment (e.g., for L-DED Inconel 718) / surface finishing (e.g., Al 6061 anodization) protocols.

Procedures, Documentation & Quality

- Extensive experience drafting in-depth protocols, technical reports, and workshop guidelines for the propulsion department.
- Facilitated knowledge transfer by documenting best practice protocols and providing historical context for key engineering decisions.
- Proficient in interpreting and applying industrial standards, including ISO 9001 (Quality), ISO 2768 (Tolerancing), ASTM E 285 80 (Ablative Testing), among others.

Propulsion & Data Analysis

- Specific expertise in hybrid rocket engine predictive modelling and

Adira Addcreative, Gaia, Portugal — *Material Engineer Intern*

February 2023 - July 2023

Bachelors dissertation, “Powder Characterization for Laser Powder Bed Fusion”

- Characterised the different material powder feedstocks for L-PBF: density, compatibility, flowability, morphology, and particle size distribution, whilst evaluating the degradation of metal powder over several prints and powder recirculations.

EDUCATION

FEUP, Porto, Portugal — *MSc Material Engineering, Metallurgy*

September 2023 - October 2025

EPMA, Lund, Sweden — *Powder Metallurgy Summer School*

June 2025

- Completed an intensive 5-day technical programme covering the full spectrum of Powder Metallurgy (PM) technologies, from powder production to post-sintering treatments.
- Acquired comprehensive knowledge of core and advanced manufacturing processes (Sintering, MIM, HIP, Additive Manufacturing) and analyzed material properties for steels, light metals, and functional alloys.

FEUP, Porto, Portugal — *BSc Material Engineering*

September 2020 - July 2023

Cambridge School, Porto, Portugal — *C1 CAE 195/210*

October 2021

PROJECTS

FEUP & Palbit, Aveiro, Portugal — *R&D Engineer Intern*

October 2024 - Jan 2025

- Topological optimization and Finite Element Analysis (FEA) of machining head cutter within nTop, manufactured by L-PBF.
- Poster presentation at Hi_rEV conference and publication of an article at Metals MDPI “Topology Optimization of a Milling Cutter Head for Additive Manufacturing” (<https://doi.org/10.3390/met15070729>).

FEUP & Hypermetal, Gaia, Portugal — *Researcher Student*

October 2023 - Jan 2024

- Characterization from feedstock to final parts of Inconel 718 produced by L-PBF
- Published a conference article at Euro PM2025, titled “Characterization Of Inconel 718 Powder And Specimens Produced By L-PBF In Different Orientations” (<https://doi.org/10.59499/ep256768283>)

FEUP, Porto, Portugal — *Researcher Student*

October 2023

- Conducted an investigation to optimize heat treatment protocol of Inconel 718 produced by L-DED.
- Published a conference article at Euro PM2025, titled “Heat Treatments Of Inconel 718 Produced By L-DED” (<https://doi.org/10.59499/ep256767779>)

understanding concepts ranging from fluid dynamics to combustion.

– Proficient in Python (Pandas, Matplotlib) for analyzing experimental data and optimizing parameters through Design of Experiments (DoE) methodologies.

Languages

Portuguese – native

English – C1 CAE –Score 195/210

Spanish – A1

French – A1

Soft Skills & Competencies

Leadership

Agile & Adaptable

Collaborative Team Spirit

Rigorous Organization

Technical Proficiency

– General: Microsoft Office, LaTeX, Github, Python, Pandas, Matplotlib, pymatgen, TSL OIM softwares, Coolprop, CEA (Chemical Equilibrium with Applications), Ces Eductack.
– Characterisation Technologies: OM, SEM, EDS, EBSD, XRD, XRF, DSC, LD, PSD, Hausner ration, Funnel Flow, Tensile test, Compression test, Fatigue test
– CAD: NX Siemens, Fusion 360, Onshape, nTop
– CAM: Powermill, Materialise Magic
– FEA: nTop, Abaqus

References

Prof^a Elsa W. Sequeiros — Master’s Dissertation Supervisor;
ews@fe.up.pt

Msc^o Afonso Santos — Ex INVICTUS Project Lead;
up202004955@up.pt

Prof^o José Costa — Bachelor’s Dissertation Supervisor;
jmmc@fe.up.pt